

WASHINGTON HARBOR, DC, USA

WMO: 997314

Lat: 38.873N

Lon: 77.021W

Elev: 33

StdP: 14.68

Time Zone: -5.00 (NAE)

Period: 08-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) 16.0	(c) 20.7	(d) N/A	(e) N/A	(f) N/A	(g) N/A	(h) N/A	(i) N/A	(j) 23.6	(k) 33.1	(l) 20.9	(m) 31.3	(n) 9.4	(o) 310	(p) 0.405

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 7	(b) 13.4	(c) 93.3	(d) N/A	(e) 90.7	(f) N/A	(g) 88.5	(h) N/A	(i) N/A	(j) N/A	(k) N/A	(l) N/A	(m) N/A	(n) N/A	(o) 7.1	(p) 320

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a) N/A	(b) N/A	(c) N/A	(d) N/A	(e) N/A	(f) N/A	(g) N/A	(h) N/A	(i) N/A	(j) N/A	(k) N/A	(l) N/A	(m) N/A	(n) N/A	(o) N/A	(p) N/A

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a) 19.4	(b) 16.9	(c) 14.7	(d) DB	(e) 13.0	(f) 96.9	(g) 5.3	(h) 3.4	(i) 9.2	(j) 99.4	(k) 6.1	(l) 101.4	(m) 3.1	(n) 103.3	(o) -0.7	(p) 105.8	(q) N/A
(4) 19.4	(5) 16.9	(6) 14.7	(7) WB	(8) N/A	(9) N/A	(10) N/A	(11) N/A	(12) N/A	(13) N/A	(14) N/A	(15) N/A	(16) N/A	(17) N/A	(18) N/A	(19) N/A	(20) N/A

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	59.2	35.5	39.6	46.5	59.1	68.9	76.4	80.8	78.4	72.5	60.7	49.2	41.3
	DBStd	17.15	8.84	9.07	8.99	7.56	7.01	5.29	4.58	3.90	5.64	7.12	7.17	7.31
	HDD50	1340	455	307	175	13	0	0	0	0	0	8	98	284
	HDD65	3836	915	711	575	207	44	2	0	0	6	168	474	734
	CDD50	4694	6	16	67	286	587	793	955	880	676	339	74	15
	CDD65	1716	0	0	2	31	166	345	490	415	232	35	0	0
	CDH74	14631	0	0	13	199	1042	2908	5218	3606	1493	151	1	0
	CDH80	4959	0	0	0	38	250	960	2141	1145	394	31	0	0
	WSAvg	6.0	6.9	6.9	6.9	6.6	5.7	5.8	5.6	5.1	5.0	5.5	5.7	6.4
	PrecAvg	40.0	2.8	2.6	3.3	2.9	3.8	3.9	4.0	3.5	3.7	3.2	3.0	3.2
(15) Precipitation	PrecMax	66.3	7.1	5.7	8.5	6.9	10.7	14.0	9.7	9.2	12.4	9.4	7.6	6.5
	PrecMin	28.4	0.4	0.4	0.0	0.0	0.7	0.9	0.7	0.4	0.1	0.0	0.3	0.4
	PrecStd	7.6	1.3	1.4	1.5	1.3	2.0	2.8	2.1	2.0	2.8	2.1	1.8	1.6
	PrecStd	7.6	1.3	1.4	1.5	1.3	2.0	2.8	2.1	2.0	2.8	2.1	1.8	1.6
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	62.5	68.5	75.6	85.3	89.5	94.8	98.0	93.7	90.6	83.7	70.4	65.0
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	2%	DB	57.6	61.7	69.3	79.5	85.7	90.8	94.7	90.5	87.7	77.9	66.3	60.2
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	5%	DB	52.8	56.2	64.8	75.3	82.6	88.1	92.0	88.3	84.7	74.4	63.2	56.9
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	10%	DB	47.8	51.9	60.2	71.7	79.8	85.5	89.4	86.0	81.8	71.1	60.2	52.2
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	2%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	5%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	10%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(35) Mean Daily Temperature Range	5% DB	MDBR	12.5	13.9	14.6	16.6	14.9	13.7	13.4	13.0	13.2	13.3	13.4	11.3
		MCDBR	19.1	20.8	21.7	21.3	17.1	16.2	15.7	15.1	15.5	17.7	18.1	17.6
		MCWBR	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	5% WB	MCDBR	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
		MCWBR	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(40) Clear-Sky Solar Irradiance	taub		0.309	0.323	0.358	0.404	0.445	0.502	0.523	0.518	0.430	0.363	0.325	0.311
	taud		2.488	2.430	2.396	2.286	2.203	2.083	2.042	2.040	2.261	2.453	2.517	2.519
	Ebn at Noon		276	286	285	277	267	251	244	241	258	268	267	267
	Edh at Noon		25	30	35	41	45	51	53	52	39	29	24	23
(44) All-Sky Solar Radiation	RadAvg		661	928	1235	1562	1761	1945	1908	1701	1386	1039	753	581
	RadStd		55	103	90	119	197	148	98	90	161	115	64	56

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (75 neighbors)	+0.74	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+223	+130

Nomenclature: See separate page